

Plan: Blog & CMS myUnila

Versi: 1.0 — 2026-05-09 **Author:** UPT TIK Universitas Lampung **Status:** Draft — Brainstorming

1. Executive Summary

myUnila sudah **70% punya CMS** (`manajemen-konten-service` Go di port 9008-an, container `myunila-man-konten-staging` running). Yang belum: **public blog frontend** (placeholder `blog-unila` masih `a_coming_soon=true`) + **migrasi dari Blogger eksternal** (currently auto-synced via `monitoring/blog_sync`).

Rekomendasi Akhir: Opsi A — Extend `man-konten` + Build Next.js Public Site (skor decision matrix 9.0/10). - Leverage backend yg sudah ada (skema profesional: slug, banner, tags, pinned, featured, view_count, expiry, target_role) - Build Next.js SSG/ISR public site `/blog` (atau subdomain `blog.myunila.ac.id`) - Tambahkan editor modern TipTap + Media Library UI - Migrasi konten Blogger (one-time) via `blog_sync` yg sudah ada

Estimasi: MVP 6-8 minggu (~2 bulan), tim 2-3 dev.

2. Status Quo

2.1 Komponen Existing yang BISA DIPAKAI

Komponen	Tech	Status	Fungsi
<code>manajemen-konten-service</code>	Go Fiber	Running	Backend CMS — modul: pengumuman, kategori, notif, upload
<code>monitoring/blog_sync</code>	Go	Running	Sync post dari Google Blogger ke <code>monitoring.blog_posts_cache</code>
<code>/dashboard/manajemen-konten</code>	Next.js	Live	Admin UI (sudah ada)
Schema <code>pengumuman</code> table	SQL Server	Tersedia	Sudah punya field profesional: tipe (pengumuman/berita/artikel), slug, banner, tags, is_pinned, is_featured, tgl_terbit, tgl_expiry, status, target_role, view_count, allow_comment
<code>blog-unila</code> placeholder	—	TBD	Public frontend belum ada (url= <code>#</code>)

2.2 Ekosistem Eksternal

- **blog.unila.ac.id** — kemungkinan running di Google Blogger (di-sync via `blog_sync`)
- **www.unila.ac.id** — homepage utama (CMS terpisah, kemungkinan WordPress lama)
- Konten editorial saat ini terfragmentasi: Blogger + WordPress + manual sosmed posts

2.3 Schema `pengumuman` (sudah ada — nilainya tinggi)

```
id_pengumuman, tipe, judul, slug, ringkasan, isi,  
id_kategori, banner_url, author, tags,  
is_pinned, is_featured,  
tgl_terbit, tgl_expiry, status,  
target_role, view_count, allow_comment
```

Schema ini **sudah cukup untuk modern blog CMS**. Tinggal tambah: - `seo_title` ,
`seo_description` , `og_image` (SEO/social share) - `reading_time` (auto-calc dari word count) -
Relasi `tag_master` (kalau mau tag autocomplete)

3. Use Cases — 3 Jenis Konten

Use Case	Audience	Channel	Tools
A. Pengumuman Internal	Mhs/Dosen/Tendik internal	Portal myUnila (login required)	<code>man-konten</code> (sudah ada)
B. Blog/Berita Publik	Calon mhs, alumni, publik	<code>blog.myunila.ac.id</code> (no login, SEO)	TBD (focus plan ini)
C. Artikel Riset/Akademik	Akademik internal & publik	Both	Hybrid

Plan ini fokus ke **Use Case B** (blog publik) — tapi pakai backend yg sama (`man-konten`) supaya editor cuma 1.

4. Referensi Industri

Platform	Tipe	Pros	Cons
WordPress.com / self-host	Open-source, PHP	Mature, banyak plugin (Yoast SEO, multilingual, image), editor WYSIWYG, ekosistem besar	Stack PHP + MySQL beda dari myUnila, attack surface, sync ke man-konten kompleks
Headless WordPress + Next.js	Hybrid	Editor familiar, frontend modern	Double maintenance
Strapi	Headless, Node.js	Free, admin UI bagus, REST/GraphQL	Tambah stack Node, perlu hosting baru
Directus	Headless, Node.js + Postgres	Postgres native, powerful	Sama, tambah stack
Sanity.io	SaaS	Fast, structured content, GROQ query	Vendor lock-in, biaya bulanan
Ghost	Open-source Node.js	Newsletter native, modern UI	Bukan general CMS, terbatas blog only
Hugo / Jekyll (SSG)	Static	Fast, SEO, gratis hosting	Editor butuh Git knowledge — tidak ramah PR officer
Custom Go (man-konten existing)	Sudah ada	Stack konsisten myUnila	Harus build editor & media UI
Google Blogger (current)	SaaS gratis	Simple, sync ke search engine	UX kuno, customization terbatas, vendor lock-in (Google)

5. Empat Opsi Strategis

5.1 Opsi A — Extend **man-konten** + **Next.js Public Site** ☒ (RECOMMENDED)

Konsep: Backend tetap pakai **man-konten** yg sudah running. Tambahkan editor modern + media library yg bagus, lalu build Next.js public frontend untuk blog SEO-friendly.

Apa yang dibangun: 1. **Tambahan kolom SEO** di tabel **pengumuman** : **seo_title** , **seo_description** , **og_image** , **reading_time** , **meta_keywords** 2. **Editor modern:** TipTap atau EditorJS di dashboard admin (Next.js sudah pakai React) 3. **Media Library UI** — drag-drop upload, crop, alt-text, kategorisasi (sudah ada modul **upload** , perlu UI) 4. **Tag autocomplete** + relasi **tag_master** 5. **Public frontend** Next.js standalone: - Route: **blog.myunila.ac.id**

atau `myunila.ac.id/blog` - SSG (Static Site Generation) untuk performa & SEO - ISR (Incremental Static Regeneration) — auto-update konten baru tanpa rebuild - Sitemap.xml, robots.txt, RSS feed, OpenGraph, JSON-LD - Search via Meilisearch (sudah ada di myUnila) 6. **Komentar** (Phase 2) — Disqus/Giscus/native 7. **Newsletter** (Phase 2) — newsletter subscriber + email cron 8. **Migration dari Blogger**: jalankan `blog_sync` → import `blog_posts_cache` ke `pengumuman` (one-time)

Pros: - Stack 100% konsisten dengan myUnila (Go + Next.js + SQL Server) - Single auth, single source of truth - 70% backend sudah jadi → time-to-market cepat - Tidak ada vendor lock-in - SEO control penuh (Next.js paling top untuk SEO)

Cons: - Harus build editor (TipTap setup ~1 minggu) - Build Media Library UI (~2 minggu) - Migrasi Blogger butuh transformation logic

Estimasi: **MVP 6-8 minggu**, full feature 12 minggu.

5.2 Opsi B — Headless WordPress + Next.js

Self-host WordPress, expose via WPGraphQL/REST, build Next.js frontend.

Pros: Content editor familiar untuk PR officer; plugin ekosistem (Yoast, ACF, dll); WordPress mobile app native. **Cons:** Tambah stack PHP+MySQL; perlu sync atau abandon `man-konten`; double maintenance; security WP terkenal banyak vuln.

Estimasi: 8-10 minggu. Skip kalau tim TIK tidak mau extra stack.

5.3 Opsi C — Headless CMS (Directus / Strapi) + Next.js

Pakai Directus (Postgres native) atau Strapi (Node.js).

Pros: Admin UI super polished out-of-the-box; modern editor; multi-content-type; revision history; role-based access bawaan. **Cons:** Tambah stack Node.js + Postgres; abandon `man-konten` yg sudah running; learning curve; data harus migrasi dari SQL Server `pengumuman` ke Postgres Directus.

Estimasi: 10-14 minggu.

5.4 Opsi D — Pure WordPress Self-Host (Status Quo Improved)

Tetap pakai WordPress (atau pindah dari Blogger ke self-host WP), tidak terintegrasi dengan myUnila.

Pros: Simple, deploy WP biasa, banyak template university gratis. **Cons:** SILO total — 2 portal terpisah, SSO sulit, double login, branding tidak konsisten, `man-konten` jadi sia-sia.

Estimasi: 4 minggu. **NOT recommended** kecuali mau total separation.

6. Decision Matrix

Kriteria	Bobot	Opsi A (Extend)	Opsi B (Headless WP)	Opsi C (Directus)	Opsi D (Pure WP)
Time to market	25%	9	6	5	8
Reuse existing (man-konten)	20%	10	4	3	1
UX seragam myUnila	15%	10	8	8	3
Editor experience	15%	7	10	10	9
Maintenance jangka panjang	10%	9	6	7	5
SEO performance	10%	10	8	9	6
Cost (lisensi + dev)	5%	10	7	8	9
TOTAL SKOR	100%	9.00	6.50	6.45	5.30

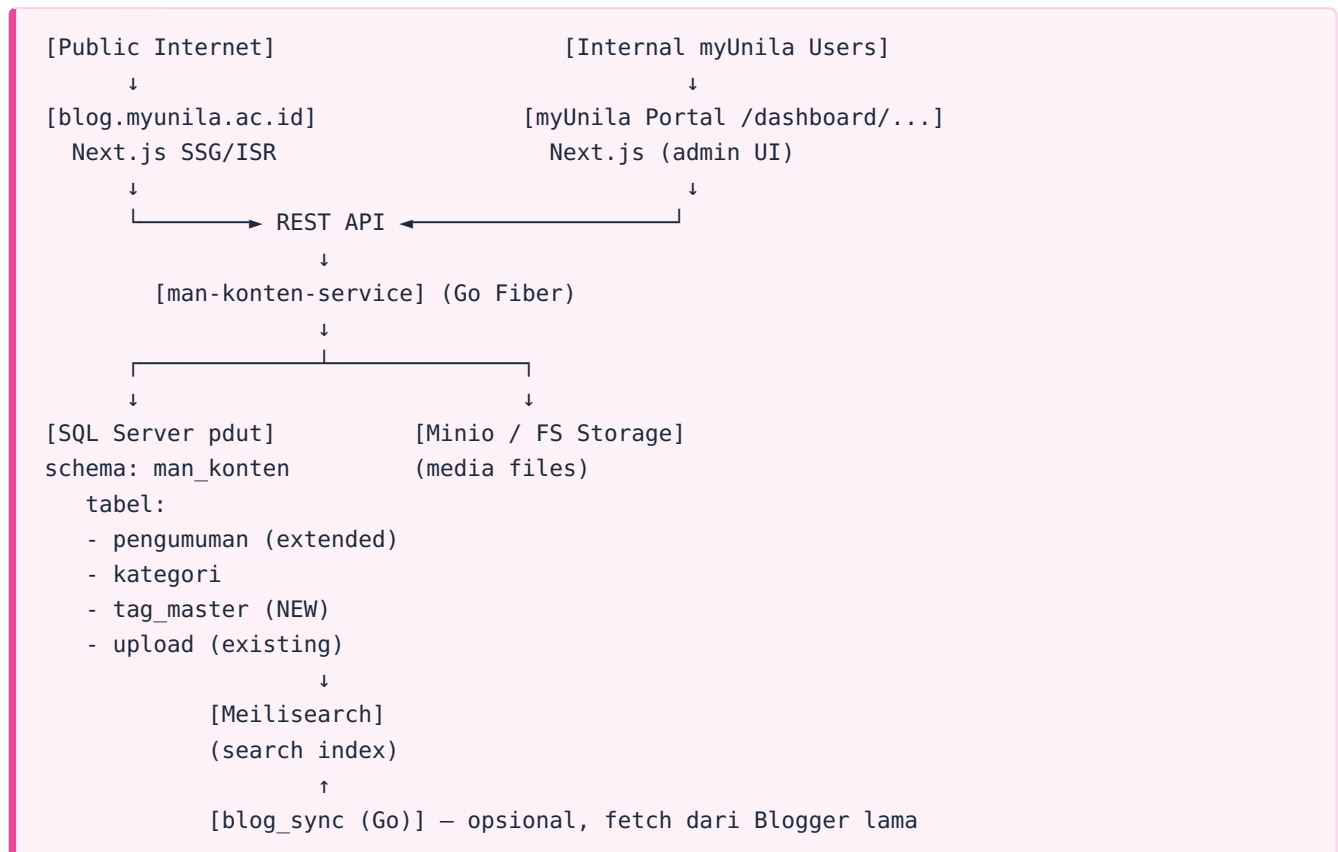
Pilihan: Opsi A — Extend man-konten + Next.js Public Site (skor 9.00/10).

7. Stack & Arsitektur (Opsi A)

7.1 Komponen

Layer	Tech	Status	Catatan
Backend CMS	man-konten-service (Go Fiber)	☐ Existing	Tambah field SEO + endpoint baru
Admin UI	Next.js 15 di myUnila	☐ Existing	Tambah TipTap editor + Media Library
Public frontend	Next.js 15 (new)	☐ Build	SSG/ISR, deploy di subdomain blog.myunila.ac.id
Database	SQL Server pdut schema man_konten	☐ Existing	Tambah kolom SEO + tabel tag_master
Search	Meilisearch	☐ Existing	Tambah indeks blog_posts
Media	Minio S3-compatible atau Filesystem + CDN	TBD	Untuk gambar/banner berita
Image Optimization	Next.js Image + sharp	☐ Existing	Auto resize/webp/lazy-load
Sync Blogger (legacy)	monitoring/blog_sync (Go)	☐ Existing	One-time import + ongoing sync (opsional)
Notif/Newsletter (Phase 2)	Email Service (myUnila SMTP)	TBD	Cron + queue
Comments (Phase 2)	Giscus / Disqus / native	TBD	Pilih saat Phase 2

7.2 Diagram



7.3 Schema Extensions

Tabel `man_konten.pengumuman` — TAMBAH kolom:

```
ALTER TABLE man_konten.pengumuman ADD
  seo_title NVARCHAR(200) NULL,
  seo_description NVARCHAR(500) NULL,
  og_image NVARCHAR(500) NULL,
  meta_keywords NVARCHAR(300) NULL,
  reading_time INT NULL DEFAULT 0,
  cover_caption NVARCHAR(255) NULL;
```

Tabel baru `man_konten.tag_master` :

```

CREATE TABLE man_konten.tag_master (
  id_tag UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
  nama VARCHAR(80) NOT NULL UNIQUE,
  slug VARCHAR(100) NOT NULL UNIQUE,
  jumlah_pakai INT DEFAULT 0,
  tgl_create DATETIME DEFAULT GETDATE()
);
CREATE TABLE man_konten.pengumuman_tag (
  id_pengumuman UNIQUEIDENTIFIER NOT NULL,
  id_tag UNIQUEIDENTIFIER NOT NULL,
  PRIMARY KEY (id_pengumuman, id_tag),
  FOREIGN KEY (id_pengumuman) REFERENCES man_konten.pengumuman(id_pengumuman),
  FOREIGN KEY (id_tag) REFERENCES man_konten.tag_master(id_tag)
);

```

Tabel baru **man_konten.newsletter_subscriber** (Phase 2):

```

CREATE TABLE man_konten.newsletter_subscriber (
  id_subscriber UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
  email VARCHAR(255) NOT NULL UNIQUE,
  nama VARCHAR(150),
  status VARCHAR(20) DEFAULT 'active',
  tgl_subscribe DATETIME DEFAULT GETDATE(),
  unsubscribe_token VARCHAR(64)
);

```

8. Feature Roadmap

8.1 MVP — Minggu 1-8

Minggu	Fitur	Owner
1	DDL extension + migrasi schema	Backend Go
1-2	Endpoint baru man-konten: tag CRUD, search by tag, related posts, sitemap data	Backend Go
2-3	Admin UI: TipTap editor (rich text + image inline), tag autocomplete, slug auto-gen	Frontend
3-4	Admin UI: Media Library page (upload, browse, alt-text, copy URL)	Frontend
4-5	Public Next.js site setup (subdomain DNS, blog.myunila.ac.id atau / blog di portal)	DevOps + Frontend
5-6	Public site pages: home (latest+featured+pinned), single post, kategori, tag, search, archive	Frontend
6-7	SEO: meta tags, OpenGraph, JSON-LD, sitemap.xml, robots.txt, RSS feed	Frontend
7	Migration script: import dari <code>blog_posts_cache</code> (Blogger) ke <code>pengumuman</code>	Backend Go
7	Meilisearch integration: index search publik	Backend
8	UAT, fix bugs, manual editor + content team	Tech Writer + QA

Output MVP: - ☐ Editor admin yg modern (TipTap) - ☐ Media library - ☐ Public blog SEO-friendly (sitemap, RSS, OG) - ☐ Migrasi dari Blogger lama - ☐ Search publik

8.2 Phase 2 — Minggu 9-16

Fitur	Detail
Komentar publik	Pilih: Giscus (GitHub), Disqus, atau native (perlu moderation)
Newsletter subscriber	Form subscribe di footer + email cron tiap minggu
Email send service	Integrate dengan SMTP myUnila / SES
Multi-author + permission	Editor / Author / Contributor roles (extend man_akses peran fungsional)
Workflow draft → review → publish	Approval workflow seperti SIMBAK
Scheduled publish	Auto-publish di tanggal tertentu (cron job)
Image CDN	CloudFlare / KeyCDN untuk image hosting
Multilingual (ID/EN)	Support translation per post
Analytics	Google Analytics 4 atau Plausible self-host
Heatmap (opsional)	Hotjar atau Microsoft Clarity (gratis)

8.3 Phase 3 — Bulan 5+ (advanced)

Fitur	Detail
AI-assisted writing	Anthropic API utk draft article, rewrite, SEO suggest
Auto-translate	English versi auto-generate via DeepL atau Anthropic
Podcast/Video	Audio embed, video player native
Author profile pages	Profile lengkap dosen/staf yg sering nulis
Related research	Auto-link artikel ke publikasi dosen di SI Penelitian
Mobile app (opsional)	Reader app pakai React Native (kalau perlu)

9. Integrasi dengan Modul myUnila Existing

Modul Existing	Integrasi Blog
manajemen-konten (existing)	Reuse backend, extend untuk public-facing
monitoring/blog_sync (existing)	One-time migration dari Blogger; ongoing sync optional
public-service (existing)	Tambah endpoint blog data ke search Meilisearch
Meilisearch (existing)	Index blog_posts baru, expose ke search bar utama myUnila
Manajemen Akses	Peran "Editor Konten" / "Author" custom (peran fungsional, bukan identitas)
SI Penelitian / SI Pengabdian / SI Publikasi	Auto-generate "Related Articles" dari publikasi dosen
Tracer Study	Embed call-to-action ke alumni di artikel relevan
LMS (future)	Embed materi blog jadi reading material di kelas
Akreditasi	Export data partisipasi konten dosen utk borang BAN-PT (publikasi populer)

10. Effort & Resource

10.1 Tim Inti

Role	Tugas	Effort MVP
Backend Go	Extend man-konten, tag CRUD, sitemap endpoint, migrasi Blogger	0.7 FTE × 8 minggu
Frontend Next.js	Editor TipTap, Media Library, public site SEO	1.0 FTE × 8 minggu
DevOps	Subdomain DNS, Minio setup, SMTP config (Phase 2)	0.3 FTE × 4 minggu
Content/Editor	Migrasi konten lama, write 10 artikel sample	0.5 FTE × 4 minggu
PM/Lead	Koordinasi, validasi PR officer	0.3 FTE × 8 minggu
QA/Tester	Cross-browser, mobile, SEO test, Lighthouse score	0.3 FTE × 2 minggu

Total: ~3 FTE selama 8 minggu MVP.

10.2 Infrastruktur

Item	Spek	Estimasi biaya
Subdomain <code>blog.myunila.ac.id</code>	DNS A record	Gratis (sudah punya)
Container Next.js public	2 vCPU, 4GB RAM	Existing VM5 atau VM baru
Storage media (Minio)	500GB awal	Existing atau ~Rp 200k/bln
CDN image (opsional)	CloudFlare free tier	Gratis (tier free cukup)
SMTP service	Existing or SES (Phase 2)	~Rp 100k/bln SES
Email subscriber list size	Up to 10k MVP, 100k+ Phase 2	(mostly free di self-host)

10.3 Budget Software

Item	Tipe	Biaya
TipTap editor	Open source MIT	Gratis
Meilisearch	Open source	Existing (gratis)
Next.js	MIT	Gratis
Image processing (sharp)	MIT	Gratis
Plagiarism check (Phase 3)	SaaS opsional	TBD
AI writing (Phase 3)	Anthropic API	Pay-per-use, ~Rp 500-1000/artikel

11. Risk & Mitigation

Risk	Probabilitas	Impact	Mitigasi
Editor lambat dgn artikel panjang	Med	Med	Lazy load images, debounce save, autosave per 30s
SEO ranking turun saat migrate dari Blogger	High	High	301 redirect tiap URL Blogger → Next.js, robots.txt, submit sitemap baru ke GSC
Spam komentar	High	Med	reCAPTCHA + moderation queue (Phase 2)
Editor PR officer kurang tech-savvy	High	Med	Training video + doc, undo/redo, autosave, paste from Word
Image storage penuh	Med	Med	Quota per author + auto-cleanup unused media >1 thn
DDoS public site	Low	High	CloudFlare proxy, rate limit Kong
GDPR/data privacy newsletter	Low	Med	Double-opt-in + unsubscribe link
Konten tidak terbaca mobile	Med	Med	Mobile-first design, Lighthouse score >90
Backup konten gagal	Low	High	Daily backup DB + media to off-site

12. Pertanyaan untuk Pimpinan

1. **Domain:** pakai `blog.myunila.ac.id` atau gabung di `myunila.ac.id/blog` ? Atau migrasi total dari `blog.unila.ac.id` ?
2. **Migrasi Blogger:** kapan deadline cutover? Perlu paralel dulu (Blogger live + new live)?
3. **Editor team:** siapa yg jadi PIC konten? Humas, PR Rektorat, atau distributed per fakultas?
4. **Branding:** pakai nama "Blog Unila" / "myUnila News" / "Unila Stories"?
5. **Multi-author:** berapa user yg perlu role Author / Editor?
6. **Hosting:** subdomain di datacenter Unila atau Cloud?
7. **Komentar:** izinkan publik komentar atau closed?
8. **Newsletter:** kirim ke email semua mhs/dosen/alumni atau opt-in saja?
9. **Multilingual:** prioritas Indonesia + English atau Indonesia saja?
10. **Akreditasi:** butuh data publikasi konten dosen utk borang? Kapan deadline?

13. Quick Win — Apa yang Bisa Dimulai Minggu Depan

- **Hari 1-2:** ALTER schema `pengumuman` tambah kolom SEO (DDL siap, idempotent)
- **Hari 3-4:** Add tag_master + relasi tag (DDL)
- **Minggu 1:** Setup Next.js public site skeleton + DNS subdomain
- **Minggu 1-2:** Editor TipTap di admin UI (research + integration)
- **Minggu 2:** Test migrasi 10 post dummy dari Blogger ke pengumuman

Setelah Quick Win, kalau approval lancar → langsung ke MVP 8 minggu.

14. Comparison Cepat: Blogger vs WordPress vs Custom (untuk pimpinan)

	Blogger	WordPress	Custom (Ops A)
Editor friendly:	****	*****	**** (TipTap modern)
SEO power:	***	*****	*****
Customization:	**	****	*****
Integration myUnila:	*	**	*****
Maintenance load:	N/A (gratis)	Medium	Low (sudah ada)
Vendor lock-in:	**** (Google)	*	*
Mobile experience:	***	****	***** (Next.js)
Cost per year:	Free	~Rp 5-10jt	Internal team

Custom (Ops A) menang di 6 dari 8 kriteria.

15. Kesimpulan

Blog/CMS myUnila layak dibangun cepat karena: - 70% backend (`man-konten-service`) sudah running, schema profesional - `blog_sync` siap migrasi konten Blogger lama - Stack myUnila (Next.js + Go + SQL Server) cukup mumpuni untuk modern blog - Kompetitor (Blogger lama) tidak SEO-friendly, susah customize

Rekomendasi: Ops A — Extend `man-konten` + Build Next.js Public Site. - MVP 8 minggu (~2 bulan) - 3 FTE - Software gratis, infra mostly existing - Time-to-market paling cepat

Setelah MVP: Phase 2 (komentar, newsletter, multilingual) 2 bulan, Phase 3 (AI, podcast, mobile) opsional.

Next Step: 1. Validasi 10 pertanyaan section 12 ke pimpinan UPT TIK + Humas 2. Eksekusi Quick Win minggu 1 3. Detailed sprint plan untuk MVP 8 minggu

Dokumen ini brainstorming hasil request Pak Mizar (lisensi@unila.ac.id) — perlu validasi & iterasi sebelum eksekusi. File: `/var/www/my-unila/docs/operations/Plan_Blog_CMS_myUnila.{md,pdf}`